

Configuration-driven experimental pipeline

one YAML defines the run; the same runner produces evaluation artifacts for the paper

one experiment = one YAML config

Backbone

Qwen2-VL, BLIP-2, PaliGemma

Dataset

DocVQA, ChartQA, ScienceQA, A-OKVQA, VQAv2

Objective

baseline, contrastive, generative, expl.-aware

Metrics

accuracy, ANLS, ROUGE/BLEU, drift

Experiment runner

builds model, dataset, objective, collator, and metrics
verifies target masking before training

Baseline eval

score the frozen model

QLoRA fine-tune

train adapter weights

Final eval

re-score the adapter

Answer

exact / multiple-choice
ANLS / relaxed accuracy

Explanation

ROUGE-L, BLEU, BERTScore

Faithfulness

attention alignment
mask → answer drift

Saved artifacts

resolved config, adapter, generations, results JSON/CSV

Paper outputs

tables, curves, transfer matrices, qualitative panels